

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2021-2022
FULL MARKS: 150

CSE 4301: Object Oriented Programming

Programmable calculators are not allowed. Do not write anything on the question paper.
 Answer all 6 (six) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. a) Write short notes on the following topics: 3 × 5
(CO1)
(PO1)
 - i. Function overloading
 - ii. Virtual Base Class
 - iii. Virtual Function
 - iv. Abstract Class
 - v. Forward Declaration
- b) Discuss the process of making an executable file for a multi-file c++ application. Discuss the difference between declaration and definition of variable, function, and class. In a multi-file application, explain the usage of declaration and definition. 10
(CO1)
(PO1)

2. In a cellular network system in figure 1, there are Base Transceiver Station (BTS) and Cellular devices (C). Multiple C objects registered under one BTS. A cellular object does not communicate directly to other cellular objects. It communicates via a BTS. The sending cellular object sends data to its registered BTS (Uplink Communication). Then that BTS will send the data to the BTS where the receiver cellular object is registered to. After that, the receiver cellular object receives data from the BTS (Downlink Communication). 30
(CO4)
(PO3)

Each BTS contains multiple Resource Block (RB) objects for both uplink and downlink cellular communication. Uplink and Downlink resources are different. Assume that BTS assigns one uplink and one downlink RB object to each registered C when it starts any cellular communication. After communication ends, BTS yields back the resources so that it might be used by another C starting cellular communication. It should be noted that a C object registered under one BTS does not mean that C has started cellular communication.

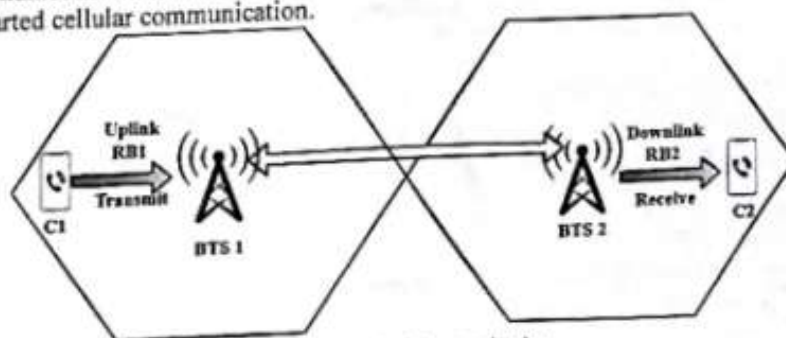


Figure 1: Cellular Communication

Write an OOP program where you can simulate the above scenario. To do so you will need class definitions of BTS, C, and RB. Use STL container instead of fixed size array where it is applicable. In the following table, some noteworthy attributes and behaviors are mentioned. You must incorporate these attributes and behaviors. You may need to add functions other than these. You are not required to write down the main() function.

You need to write exception classes also to throw specific exceptions. In the following table 1, scenarios are mentioned where the exceptions should be raised.

Table 1: Attributes, behavior and exception for Question 2

Class Name	Attributes	Behavior	Exception
BTS	2D Coordinate, uniqueID	allocateRBCell(RB*,C*) deallocateRBCell(RB*,C*) getFreeRBs() - return a vector of free RBs under that BTS	1. Trying to allocate already allocated RBs. 2. Trying to allocate another RB to a Cellular Device which is already using RB. 3. A cellular device want to start the communication but no free RBs are present
C	2D Coordinate, currentTransmissionPower, minTransmissionPower, maxTransmissionPower, uniqueID	startCommunication() endCommunication() status() - this function will print the uniqueID of C. If it is under cellular communication then it shows the uniqueID of RBs of uplink and downlink.	
RB	currentTransmissionPower, minTransmissionPower, maxTransmissionPower, isUplinkRB, channelFrequency, uniqueID	status() - this function will print the uniqueID of RB. If downlink RB then print minTransmissionPower, maxTransmissionPower	

3. CourseResult class is responsible for storing the result of a student for an individual course. It stores the information of Course Code, Course Title, Credit, Grade Point (result of the student), and Semester. Student class is responsible for storing the information of a student. It stores the information on student name and student ID. Moreover, a student can register for multiple courses. Write necessary class definition for this scenario.

30
(CO4)
(PO3)

Write a function generateSemesterWiseResult(), which will print the result of a student. In the printed result, grades of each semester will be shown in ascending order of semester number. In the result of each semester, Grade Points in each course of that semester will be shown followed by Grade Point Average of that semester, and Cumulative CGPA earned at the end of that semester.

ResultService class is responsible for the result of all students. Create a member function sortStudents(), which sorts the Student class objects according to CGPA.

You need to include the file read and write operations of student class, so that the information is not erased when the program is closed. When the program restarts it should load the data from the file.

4. a) Consider Code Snippet in figure 2. Now overload << operator to set salary for an employee. Also overload >> operator to get the salary for an Employee.

10
(CO2)
(PO2)

```
#include <iostream>
using namespace std;

class Employee{
    int salary, bonus;
public:
    Employee(int _salary):salary(_salary),bonus(salary*10/100){}
    Employee(int _salary, _bonus):salary(_salary),bonus(salary*_bonus/10){}
};
```

Figure 2: Code Snippet for Question 4. a)

- b) Assume that you create a class by deriving from a base class which contains only a parameterized constructor. Do you need to write a constructor in the derived class? Give reasons to support your answer with example code.

5
(CO1)
(PO1)

5. a) You want to simulate the application of heat on different substances at different states of matter. The amount of heat that is required to increase the temperature of unit mass of that substance at some state by one degree is known as specific heat of that substance at that state. The temperature at which a substance's state changes from solid to liquid or vice-versa is known as melting point. On the other hand, the temperature at which a substance's state changes from liquid to gaseous or vice-versa is known as boiling point. At melting point or boiling point if you apply heat that energy is used to change the state of the substance instead of changing the temperature of the substance. The heat absorbed by a unit mass of a given solid at its melting point that completely converts the solid to a liquid at the same temperature is known as heat of fusion. The heat absorbed by a unit mass of a given liquid at its boiling point that completely converts the liquid to a gaseous state at the same temperature is known as heat of vaporization.

13
(CO2)
(PO2)

If you apply heat to a solid state of a substance based on the specific heat and mass of the substance the temperature will continuously rise to melting point. Then it would take heat of fusion to convert from unit mass of solid to liquid state and there will be no change in temperature. After that, if you apply heat to a liquid state of a substance based on the specific heat of the liquid and mass of the substance, the temperature will continuously rise to boiling point. Similarly, it would take heat of vaporization to convert from unit mass of liquid to gaseous state and there will be no change in temperature.

Now create a substance class which follows the above-mentioned rules for application of heat. Sample public member functions of the class can be:

- `getTemperature()` returns the current temperature of the substance
- `getState()` returns the current state of the substance
- `applyHeat(double amount)` and `removeHeat(double amount)` change the temperature and the state of the substance accordingly

It should be noted that some of the attributes are constant which implies that it cannot be changed once you initialized. Use `const` modifier for those attributes.

- b) You created a class substance in question 5. a). Now assume that you want to simulate the mixing of two substance objects. It should be noted that heat transfers from high temperature to low temperature. Write a function and place it in a suitable place to simulate this behavior.
- c) Why do programmers opt for a new mechanism (known as "exception") to handle error instead of using simple if else block.

7
(CO2)
(PO2)
5
(CO1)
(PO1)

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Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. Assume that you are asked to develop a portion of a program called Course Management System for the CSE Department of IUT that considers the following requirements. 30
(CO4)
(PO3)

Teachers are assigned to different courses in an academic semester. A course has attributes like course title, course code, effective contact hour, number of sections/groups, etc. Here, effective contact hours indicate the amount of contact hours counted for a teacher in his teaching load if the course is assigned to the teacher for one section. There are three categories of courses: Theory, Lab, and Thesis/Project. The number of teachers assigned to a course is based on this category. For a theory and thesis/project course, a section is assigned to a single teacher. For a lab course, a section is assigned to two teachers. For analyzing course related requirements in the future, unique functionalities may be added for each category of courses.

A teacher has attributes like Name, Designation, Minimum Contact Hours, etc. The sum of the effective contact hours of assigned courses of each teacher (also known as load) should be equal to or greater than the minimum contact hours.

You will need a separate class that deals with the assignment of courses to teachers. This class needs to provide the following functionalities by creating one or more public methods:

- Assign a course object to a teacher object. It should be noted that different teacher objects can be assigned to a single course. For example, assume that a theory course, CSE 4301 has 2 sections. Two teachers can be assigned to this course, each of whom is assigned to one section. You can assign one teacher to both sections as well. You cannot assign a course to a teacher if all sections of that course are already assigned.
- Detach a course object from a teacher object. It should remove the assignment of the teacher to that course. Partial detachment is possible, meaning not all sections of a course will be detached from that teacher.
- Display all the teachers whose load is lower than the minimum contact hour.
- Display all the courses that are not fully assigned, indicating the number of sections yet to be assigned.
- Display the current assignment of courses to teachers.

Identify different classes from the scenario and write their definition. You can add necessary member functions and variables that are not explicitly mentioned in the scenario. Consider adding the `const` modifier to any member function if necessary.

2. Banks have many different types of accounts, often with different rules for fees associated with minimum balance, withdrawals, etc. Customers are allowed to perform three categories of transactions on their accounts: deposit, withdraw funds, and transfer funds between accounts incurring the appropriate fees associated with the withdrawal of funds from one account. It should be noted that no transaction with a negative amount is allowed and the balance of any account cannot be negative at any moment. If the balance is negative due to application of fees by the Bank, then the account is frozen. No transaction is allowed on a frozen account. A transfer transaction from the first account to the second account will count as a withdrawal from the first account and deposit at the second account while calculating any fees for an account (although it will be a transfer transaction).

Banks have two categories/types of accounts. They are current account and savings account. In case of a current account, the account holder can deposit and withdraw money without any restriction. These accounts will have a yearly account maintenance fee of BDT 1000.

On the other hand, there is a daily limit on the number of free withdrawals and the number of free deposits for each savings account. It will be different for each savings account and initialized when the account is created (the default value for each is 5). There is also a daily limit on the amount of withdrawal (no limit on deposit). It will be different for each savings account and initialized when the account is created (the default value is BDT 10,000). If the daily limit is crossed, then a fee of BDT 100 or 2% of the transacted amount (whichever is higher) will be applied.

All categories of accounts have an account holder(s) name, an account number, and a list of transactions associated with that account. One can perform functions like deposit, withdrawal, and transfer of money on any account following the policy mentioned above. It should be noted that transaction history associated with the account will not be removed from the bank even if the account is removed.

A bank has a name, a list of bank accounts, and a list of transactions. The functionalities of a bank are clearing the daily limit on all accounts (it will reset the count of daily transactions to zero for all accounts of that bank), printing all transaction histories of that bank, and printing the current balance of all bank accounts. It should be noted that all transactions of a bank will be removed if that bank is removed. All the transactions should be executed using the bank object.

Design a program for the above-mentioned scenario using object-oriented features such as encapsulation, abstraction, and polymorphism. Use an STL container to store a list.

3. a) What is friend class and friend function? What are the pitfalls of using the friend function or friend class in your program?
- b) Should one declare the destructor function as virtual if s/he plans to use the class as an abstraction for other classes? Explain your opinion with an appropriate example.
- c) Can a single catch block handle multiple exception class objects? If so, give an example code to handle multiple exceptions under one try block.
- d) Give an example code demonstrating the single accessor function of a member variable, which can both read and write.
- e) What is the necessity of a pure virtual function compared to a virtual function? Explain it with an example code.
- f) What is the use of namespace? How can different identifiers (variables, functions, or classes) be present in different files be under the same namespace? For example: `cout` object, `min` function, `string` class, etc. are defined under different classes in different files but present in the `std` namespace.

Complete the definition of the classes according to the instructions given as comments in the Code Snippet 1. You cannot use any library other than `iostream`. The output of the given `main()` is (A: 2), (B: 1).

```
1 #include <iostream>
2 using namespace std;
3 class element {
4     char key; int count;
5 public: /*Implement appropriate constructors, getter and setters*/
6 };
7 class dictionary {
8     /*Declare an object pointer of 'element' class and *other properties
9     as necessary*/
10 public: /*Create necessary constructors*/
11     /*Check if element with the given key exists*/
12     bool hasElement(char key);
13     /* Add a new element to the array of 'element' objects. Return false
14     * if element already exists. Else return true.*/
15     bool addElement(element elm);
16     /* Add a new element to the array of 'element' objects initialized
17     key, count.
18     * Return false if element already exists. Else return true.*/
19     bool addElement(char key, int count);
20     /* Remove element from array of 'element' objects. Return false if no
21     * such element exists. Else return true.*/
22     bool removeElement(element elm);
23     /* Remove element from array of 'element' objects by matching the
24     * key. Return false if no such element exists. Else return true.*/
25     bool removeElement(char key);
26     /* Increase count of the element that matches the key. Return false
27     * if no such element exists. Else return true.*/
28     bool increaseCount(char key);
29     /* Decrease count of the element that matches the key. Return false
30     * if no such element exists. Else return true. If element's count
31     * decreases to zero, remove the particular element*/
32     bool decreaseCount(char key);
33     /*Print all existing elements along with their count.*/
34     void printElements();
35 };
36
37 int main()
38 {
39     char addingList[] = "ABACB";
40     char deleteList[] = "BCCGG";
41     dictionary dict;
42
43     for (int i=0; i<5; i++)
44         if (!dict.addElement(addingList[i], 1))
45             dict.increaseCount(addingList[i]);
46     for (int i=0; i<5; i++)
47         dict.decreaseCount(deleteList[i]);
48     dict.printElements();
49 }
```

Code Snippet 1: An incomplete C++ program for Question 4.

5. a) A rational number is a number that can be expressed as the quotient or fraction p/q of two integers, a numerator p , and a non-zero denominator q . Such as $3/7$, $-5/2$ etc.

10 +
8 + 7
(CO2)
(PO2)

Create a class `RationalNumber` that implements the following member functions. Write a driver function to test the member function.

- `bool assign(int numerator, int denominator)`: it sets the value to the data member and returns `true`. Do not set the value if the denominator is 0 and return `false`.
- `double convert()`: it returns the decimal equivalent. For example: We know that $3/2=1.5$. So if the numerator and denominator are 3 and 2, respectively then, return 1.5.
- `bool invert()`: it swaps the numerator and denominator. If the `RationalNumber` object stores $3/2$, after calling this function the same object will store $2/3$. It will return `true` for successful inversion and `false` otherwise.
- Copy Constructor function and Assignment Operator Overloading function.
- Overload the `+`, `-`, `*`, `/` operators and implement them as member function. These functions should return a `RationalNumber` object.
- Overload the insertion and extraction operator so that the standard input and output stream object can be used on the `RationalNumber` object. Example: if `obj` is an object of class `RationalNumber`, `cin` and `cout` can be used on `obj` to take input and print on console, respectively.

- b) Refactor the solution code for Question 5.a) to include an exception handling mechanism. The exception occurs if the denominator is zero. Change the class definition in appropriate places so that it can raise the exception. Rewrite the driver code to handle the exception. For demonstration purposes, show an error message as a handling mechanism.

- c) Add two member functions (`readFromFile` and `writeOnFile`) that can read from the file and append the `RationalNumber` object on a file named `rational.txt`. Note that the file contains multiple `RationalNumber` objects. The `readFromFile` function will take a single parameter n to read n^{th} input from the file.

6. a) What are the advantages of compiling `.cpp` files separately?

2 +

- b) Assume that a project has classes A, B and C. For each class, a header file containing the class definition and a `.cpp` file containing the implementation of the member function is present. This project also contains `X.lib`, `X.obj`, and `X.h` from a different vendor. Describe the different steps to convert this project into an executable file.

8 + 5
(CO1)
(PO1)

- c) How to use global variables, functions, and classes that are defined in a different file? How to restrict the visibility scope of a global variable to a file itself.